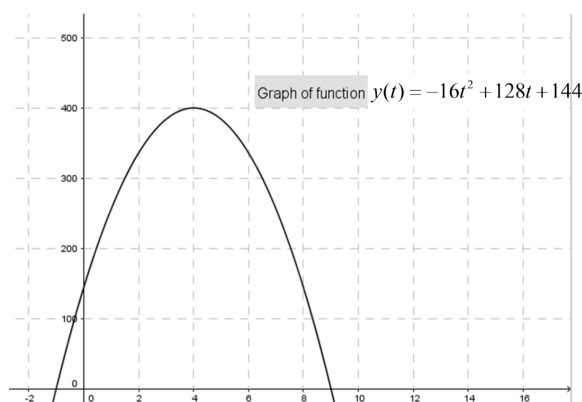


Problem 2

Problem 1 Part of the given parabola can be used to model the “position-time” graph of a ball thrown straight up into the air. The graph gives the height of the ball in feet t seconds after being thrown into the air.

Let the function f be defined by $f(t) = -16t^2 + 128t + 144$.

Use this graph, and the given function, f , to answer the following questions.



- Mark the part of the parabola that can be used to model the position of the ball.
- What are the units on the t axis? What are the units on the y axis?
- If you were watching a movie of the ball being thrown, is the graph a picture of the path that the ball follows? Why or why not?
- Let $f(t)$ denote the height of the ball at time $t, t \geq 0$. What is the height of the ball at time $t = 0$?
- When will the ball hit the ground?
- What is the domain of the position function, f , of the ball?
- Use the table of values to find the average velocity of the ball between $t = 8.9$ and $t = 9$ seconds.

t	$\approx f(t)$
8.9	15.84
8.99	1.6
8.999	0.159984
8.9999	0.015998
9	0

Problem 2

- (h) Use the table of average velocities to approximate the instantaneous velocity of the ball when it hits the ground.

Time Interval	Average Velocity
$[8.9, 9]$	$\frac{f(9) - f(8.9)}{.1} = \frac{0 - 15.84}{.1} = -158.4$
$[8.99, 9]$	$\frac{f(9) - f(8.99)}{.01} = \frac{0 - 1.5984}{.01} = -159.84$
$[8.999, 9]$	$\frac{f(9) - f(8.999)}{.001} = \frac{0 - .159984}{.001} = -159.984$
$[8.9999, 9]$	$\frac{f(9) - f(8.9999)}{.0001} = \frac{0 - .0159998}{.0001} = -159.998$

- (i) Compute $v_{AV(t)}$ the average velocity of the ball on the time interval $[t, 9]$, where $t < 9$.
- (j) Compute $v(9)$, the **instantaneous** velocity of the ball at $t = 9$.